

Academia

“Not Only Do You Get Versatile And Affordable Hardware, But You Also Gain Access To A Vibrant And Supportive Community”

What does it take to turn an idea into a working IoT solution? How can academic programmes give students hands-on skills that industry actually values? Vidushi Saxena spoke with Dr Umesh Dutta to uncover how Arduino and Manav Rachna are bridging the gap between learning and real-world innovation.

DR UMESH DUTTA
CEO, MANAV RACHNA INNOVATION
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How did Manav Rachna decide to collaborate with Arduino Labs?

Arduino has over 36 million global users and is a leader in technology, supporting schools, higher education, and industry in innovation and prototyping. When Arduino planned its India expansion, Manav Rachna aligned with the initiative to advance education, skills development, and sustainability. The partnership aims to train over 10,000 teachers and cultivate the mindset and ecosystem necessary for success.

Who supported or funded the setup for this lab?

It is a joint collaboration between Arduino and Manav Rachna, with valuable input from both organisations. It can be described as a joint investment, as both contributed to developing the course.

What would be the estimated cost of establishing such a lab, and how sustainable would its operations be in the long term?

The cost of an Arduino lab

depends on the equipment and the type of institution. The collaboration follows a three-tier ecosystem: Inspiration for schools, Invention for universities and colleges, and Innovation for industry R&D hubs. Typical setups start at ₹6-8 lakh (600,000 to 800,000 rupees), with costs varying based on hardware and customisation.

Arduino Labs emphasise hands-on, project-based learning, allowing students to experiment with hardware, write programs, and learn by doing. Students can also